

Research paper

Predefined-time stability and its applications using non-singular sliding mode control

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ARTICLE INFO

Keywords:

Predefined-time stability (P_dTS)
Lyapunov stability
Non-singular predefined-time terminal sliding mode control
Synchronization
Dynamical systems

ABSTRACT

In this work, a Lyapunov-like sufficient condition for predefined-time stability (P_dTS) is presented for a class of dynamical systems. In addition, the proposed Lyapunov-like condition allows us to deduce some prior results on P_dTS . Based on the proposed scheme, non-singular predefined-time controllers are designed for first-order systems. Further, a predefined-time terminal sliding surface (P_dTSS) and corresponding non-singular predefined-time terminal sliding mode control (NP_dTSM) strategy have been proposed for the second-order systems. These controllers have several advantages: (i) The control scheme can withstand the effects of bounded, non-vanishing disturbances and is non-singular in the whole state space. (ii) The convergence time can be specified by a constant defined by the user in advance. Along with the predefined-time synchronization of the Φ^6 duffing oscillator, a two-dimensional chaotic system, some examples are provided to justify the numerical inferences of the proposed scheme.

1. Introduction

Research on dynamical systems nowadays is primarily concerned with achieving a certain response time. Hence, the concept of finite-time stability (FTS) has come into existence recently. It has the upper hand over asymptotic and exponential stability concepts due to the convergence of the state trajectories to the desired position in some finite time. Initial works on this topic can be found in [1–3]. Further, because of its certain response time, it became hugely attractive in the field of engineering [4], including information processing [5], secure communication [6], cryptography [7] and in many biological, chemical and physical systems. However, FTS suffers from the drawback of dependency of the settling-time on the initial states of the system, which may not always be available in real-life applications. Thus, one will obtain different settling-times for a system corresponding to different initial conditions. To overcome the dependency of settling-time on initial states and to maintain uniformity among settling-time estimates corresponding to different initial values, the concept of fixed-time stability ($FxTS$) was introduced by Polyakov [8]. $FxTS$ can, therefore, be considered as a particular case of FTS where a constant bounds the settling-time and is independent of the initial conditions. Since then, researchers have paid considerable attention to fixed-time stability analysis, including [9,10]. Specifically, authors in [9] provide a new sufficient Lyapunov condition by incorporating a linear term in the Lyapunov inequality provided in [8], whereas, in [10], a novel robust fixed-time stability criterion is provided using constant and variable exponent coefficients. Despite of the noteworthy superiority of $FxTS$ over FTS , it suffers from a severe drawback, namely the difficulty of obtaining a simple relation between tuning gains and the upper bound of the settling-time. Thus, a conservative estimate of the settling time is often obtained, reducing the accuracy. For instance, in the work [11], the theoretical upper bound of the settling-time is 100 times greater than that of the actual stabilization time. Further, in many practical applications, one needs to tune the

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settling-time as desired, which is not always possible in $FxTS$. To resolve the problems described above, a new notion of stability known as predefined-time stability (P_dTS) has been introduced in [12]. P_dTS can be considered as a particular case of $FxTS$ in which settling-time's upper bound is regarded as a tunable parameter and thus can be chosen as desired. Due to the advantages mentioned above, P_dTS has attracted plenty of researchers recently (see the work discussed in [13–22]). For instance, [14] proposes a sufficient condition for autonomous non-linear dynamical systems and discusses its benefits over $FxTS$. Unlike the previous ones, where exponential terms were used for P_dTS of dynamical systems, [17] provides a new sufficient condition for P_dTS constituting polynomial terms. The authors in [21] give an adequate condition for P_dTS of dynamical systems, and based on that, they have constructed predefined time controllers for first- and second-order systems. [23] provides a P_dTS criterion for impulsive systems by considering hybrid impulses. Further, a new Lyapunov-like condition for P_dTS is provided in [19], which generalizes many of the previously constructed sufficient conditions on P_dTS of autonomous dynamical systems. Moreover, having as few as possible Lyapunov-like conditions instead of several will reduce the workload of studying each separately. Thus motivated by the above discussions, the authors in the present paper try to provide a more generalized version of the Lyapunov-like condition, which will consider many of the previously defined P_dTS results on autonomous systems.

Moreover, since the discovery of sliding mode control (SMC) by Utkin in [24], it has been recognized as the most preferred controller among researchers due to its advantages, like robustness to external disturbances, order reduction, simplicity of construction and implementation, over several control strategies like backstepping control [25], adaptive control [26], saturation control [27], PID control [28], etc. Further studies on SMC can be found in [29,30]. The implementation of SMC consists of two parts. The first is to design a sliding manifold that specifies a desired convergence rate. The second one is to construct a suitable sliding mode controller to force the state trajectories into the sliding manifold and make them remain there afterwards. Despite several advantages, the convergence rate of conventional $SMCs$ used to be asymptotic, and hence, a modified SMC known as terminal sliding mode control ($TSMC$) was introduced in [31] to obtain finite-time convergence of the state trajectories. In addition, due to different kinds of modified notions of stabilities, various forms of terminal sliding modes in the likes of finite-time, fixed-time, and predefined-time sliding mode surfaces have attracted a lot of researchers recently. In [32], the authors provided a finite-time sliding surface, whereas a fixed-time sliding surface is discussed in [8]. Recently, in [21], authors provided a predefined-time sliding surface using the approach of the fixed-time sliding surface of [8]. Besides all of these, a singularity problem generally arises in many of the control designs. An initial approach to overcome the problem of singularity in TSM was proposed in [33]. After that, several works have been done to avoid the singularity problems [34–36]. For instance, [36] proposes a methodology to overcome the singularity problem in $TSMC$, which provides finite-time convergence, whereas [34] proposes a novel non-singular terminal sliding surface and shows its uses in finite-time robust stabilization of second-order non-linear dynamical systems in the presence of matched disturbances. Recently, in [37], a non-singular predefined-time integral sliding surface is constructed. And in [38], the author uses κ^1 type function to build a predefined-time sliding surface. However, the trajectories here converge to a small region rather than converge to the origin. Further, as per the authors' knowledge, there are very few, if at all exist, P_dTSS using κ^1 functions present in the literature, and thus, authors have found that constructing P_dTSS and corresponding non-singular controllers to stabilize non-linear autonomous dynamical systems is worth studying.

Meanwhile, the synchronization of dynamical systems is an excellent application of the results on stability because the synchronization of nonlinear systems can be converted into the stability of the error system of the drive and response systems. Several works on predefined time synchronization of dynamical systems have been put forward in [39–42]. For instance, the authors of [43] have studied predefined-time synchronization of chaotic systems of different dimensions, whereas [40] studies predefined-time synchronization of delayed memristive neural networks by considering external disturbances. Predefined-time synchronization of Fuzzy cellular neural networks with stochastic perturbations is studied in [42], and Predefined-time synchronization of two different fractional-order time-delay chaotic systems is presented in [39].

Further, Chaos is a delightful but complex non-linear phenomenon we encounter in day-to-day life. Sensitive dependency of state trajectories on initial conditions, i.e., a slight deviation of initial conditions may cause a significant divergence in trajectories in the future, is one of the fundamental properties of chaotic systems. Controlling, specifically synchronizing, chaotic systems became the centre of active research after the pioneering work of Pecora and Carroll [44], who showed mathematically that chaos control is possible. The field of synchronization of chaotic systems has grown by leaps and bounds due to its vast applications in several areas, such as living systems [45], optical cryptography [46] and secure communication [47]. Also, disturbances are inevitable in real-world scenarios, and thus, synchronizing chaotic systems by considering disturbances is a challenging task. As far as the authors' knowledge goes, predefined-time synchronization of chaotic systems by considering disturbances is rarely studied in the literature. Meanwhile, SMC is an effective control strategy to cope with bounded disturbances. Thus, predefined-time synchronization of chaotic systems, constituting disturbances, using terminal sliding mode control is found to be worth studying.

In the recent past Lyapunov tools have been considered as the central strategy for stability analysis and designing of the non-linear control systems [48,49]. It has also been widely utilized to evaluate or estimate the convergence rate in systems that demonstrate finite-time [3] and/or fixed-time [8,9] stability. The situation has not varied in the case of P_dTS also.

Motivated by forgoing discussion the most important contributions of the present work can be summed up as follows:

1. Using a special kind of function known as κ^1 function, a Lyapunov-like theorem on P_dTS of autonomous dynamical systems is put forward.
2. The proposed theorem is a generalization for many of the previously predefined-time stability theorems discussed in [19,20,39,40] and hence, reduces the burden of analysing them separately.

3. For the predefined-time stability of first-order disturbed systems, a non-singular predefined-time controller is constructed. Additionally, a $P_D TSS$ and corresponding non-singular predefined-time controllers are designed to stabilize second-order systems with matched disturbances in predefined-time.
4. Numerous examples of numerical simulations are presented to demonstrate the reliability and viability of the control criteria, including synchronization of the Φ^6 duffing oscillator.

The remainder of the paper is assembled in the following way: Section 2 constitutes some mathematical preliminaries, mainly; some basic definitions and an existing Lemma on predefined-time stability. In Section 3, the main result of the paper, a novel Lyapunov-like theorem for $P_D TSS$, is provided. Moreover, this section also demonstrates how it can be considered as a generalization of existing results in literature on $P_D TSS$. Section 4 is dedicated to the applications of the present results, which consist of designing robust controllers for first and second-order systems and construction of sliding surfaces for second-order system. Additionally, predefined-time synchronization of the Φ^6 duffing oscillator is also demonstrated in this section. Finally, the major conclusive statements are highlighted in Section 5.

2. Preliminaries

This section contains some mathematical preliminaries needed for the rest of the article such as, a variety of notations, definitions of different types of stability, a special class of function and a lemma.

2.1. Notation

- $\mathbb{R}$ is the set of real numbers; $\mathbb{R}_+ = \{z \in \mathbb{R} : z > 0\}$; $\mathbb{R}_{\geq 0} = \mathbb{R}_+ \cup \{0\}$; $\bar{\mathbb{R}}_+ = \mathbb{R}_+ \cup \{\infty\}$.
- For any real number m , define $[\cdot]^m : \mathbb{R} \rightarrow \mathbb{R}$ as $[z]^m = |z|^m \text{sign}(z)$ for any $z \in \mathbb{R} \setminus \{0\}$. Furthermore, if $m > 0$, $[0]^m = 0$.
- Consider the functions $Y : \mathbb{R} \rightarrow \mathbb{R}^p$ and $\phi : \mathbb{R} \rightarrow \mathbb{R}$. Then $\dot{Y} = \frac{dY}{dt}$ indicates the first derivative of the function Y with respect to time t ; $\phi'(z)$, $\phi''(z)$ represents the first and second order derivative of the function ϕ with respect to z , where z is not the time variable t .
- For $z \in \mathbb{R}^p$, z^T denotes its transpose and $\|z\| = \sqrt{z^T z}$.
- For $\alpha_0, \beta_0 > 0$, the Gamma Functions and Beta Functions are defined as $\Gamma(\alpha_0) = \int_0^\infty z^{\alpha_0-1} e^{-z} dz$ and $B(\alpha_0, \beta_0) = \int_0^1 z^{\alpha_0-1} (1-z)^{\beta_0-1} dz$, respectively. Further, for $r \in \mathbb{R}_{\geq 0}$ define Incomplete Gamma Function and the regularized Incomplete Gamma Function as $\gamma(\alpha_0, r) = \int_0^r z^{\alpha_0-1} e^{-z} dz$ and $P(\alpha_0, r) = \frac{\gamma(\alpha_0, r)}{\Gamma(\alpha_0)}$, respectively. Eventually for $0 \leq r \leq 1$, define Incomplete Beta Function as $b(\alpha_0, \beta_0, r) = \int_0^r z^{\alpha_0-1} (1-z)^{\beta_0-1} dz$ and regularized Incomplete Beta Function as $I(\alpha_0, \beta_0, r) = \frac{b(\alpha_0, \beta_0, r)}{B(\alpha_0, \beta_0)}$.
- $B(\alpha_0, \beta_0) = \frac{\Gamma(\alpha_0)\Gamma(\beta_0)}{\Gamma(\alpha_0+\beta_0)}$, and $\Gamma(\alpha_0 + 1) = \alpha_0 \Gamma(\alpha_0)$ for $\alpha_0, \beta_0 > 0$.
- Hypergeometric Function is defined as, ${}_2F_1(\alpha_0, \beta_0; \gamma_0; u) = \frac{1}{B(\beta_0, \gamma_0 - \beta_0)} \int_0^1 t^{\beta_0-1} (1-t)^{\gamma_0-\beta_0-1} (1-tu)^{-\alpha_0} dt$, where, $\gamma_0 > \beta_0 > 0$.

2.2. On different kinds of stability

Let us look at the autonomous system:

$$\dot{z} = h(z; \sigma), z(0) = z_0 \quad (1)$$

where $z \in \mathbb{R}^p$ is the system state; the vector $\sigma \in \mathbb{R}^q$ illustrates tunable parameters of the system (1) and the function $h : \mathbb{R}^p \rightarrow \mathbb{R}^p$ may be discontinuous. The solution $\varphi(t, z_0)$ of (1) and its uniqueness is understood in the sense of Filippov [50]. Further, the origin $z = 0$ is the equilibrium point of (1).

All the stability notations defined and referred to hereafter are global, so it is skipped to indicate them further.

Definition 1 (Stability Notations [19]). The origin of system (1) is said to be,

- **Lyapunov stable:** if for any $z_0 \in \mathbb{R}^p$ and for all $\epsilon > 0$, there is $\delta(\epsilon) > 0$ such that for any $\|z_0\| < \delta(\epsilon)$, the solution $\varphi(t, z_0)$ is defined for all $t \geq 0$, and $\|\varphi(t, z_0)\| < \epsilon$ for all $t \geq 0$.
- **Finite-time stable:** if it is *Lyapunov stable* and for any $z_0 \in \mathbb{R}^p$ there exist $0 \leq \tau(z_0) < \infty$ such that the solution $\varphi(t, z_0) = 0$ for all $t \geq \tau(z_0)$. Further, the function $T(z_0) = \inf \{\tau : \varphi(t, z_0) = 0, \forall t \geq \tau(z_0)\}$ is known as the *settling-time function*.
- **Fixed-time stable:** if it is *finite-time stable* and the settling-time function $T(z_0)$ is bounded on $\mathbb{R}^p$, i.e., $\exists T_{max} > 0 : T(z_0) \leq T_{max} < \infty$ for all $z_0 \in \mathbb{R}^p$.

Remark 1. It is clearly observable that the bound T_{max} of the settling-time function is not unique. For example if $\forall z_0, T(z_0) \leq T_{max}$ then $\forall z_0, T(z_0) \leq \Lambda T_{max}$ with $\Lambda \geq 1$.

Example 1. Based on [19], consider the system

$$\dot{z} = \left(-p_0 [z]^{p_0} - p_0 [z]^{3-2p_0} \right) \quad (2)$$

where $z \in \mathbb{R}$ is the system state, $p_0 \in \mathbb{R}$ is the parameter of (2) which satisfies $0 < p_0 < 1$. By [8] it can be easily shown that the origin of (2) is fixed-time stable. Further, by [21] the settling-time function satisfies

$$\sup_{z_0 \in \mathbb{R}} T(z_0) = \frac{\Gamma(1/3) \Gamma(2/3)}{3p_0(1-p_0)} = \frac{2\pi}{3\sqrt{3}p_0(1-p_0)} > \frac{2\pi}{3\sqrt{3}}.$$

Remark 2. From the above example it can be observed that, regardless of how the parameters are set, the convergence time of the system (2), which is $> \frac{2\pi}{3\sqrt{3}}$, cannot be diminished arbitrarily.

To overcome the demerits as mentioned in Remarks 1 and 2, the notion of P_DTS was established in [12], where upper bound of the settling-time function $T_c = T_c(\sigma) > 0$ is set in advance by the user.

Definition 2 (Predefined-Time Stability [19]). The origin of system (1) is said to be P_DTS if it is *fixed-time stable* and for any user defined constant $T_c := T_c(\sigma) > 0$, there exist some $\sigma \in \mathbb{R}^q$ such that the settling-time function of system (1) satisfies

$$T(z_0) \leq T_c, \forall z_0 \in \mathbb{R}^p.$$

In this case, T_c is known as *predefined-time*. Additionally, if $\sup_{z_0 \in \mathbb{R}^p} T(z_0) = T_c$, then T_c is called the *strong predefined-time*.

2.3. Class of κ^1 functions

Definition 3 (κ^1 Functions [19]). Consider a continuous function $\xi : \mathbb{R}_{\geq 0} \rightarrow [0, 1]$ that is strictly increasing. Further, if $\xi(0) = 0$ and $\xi(r_0) \rightarrow 1$ as $r_0 \rightarrow \infty$ holds, the function ξ is called a κ^1 function and it is then denoted by $\xi \in \kappa^1$.

Remark 3. [19] It is to be noted that if $\xi \in \kappa^1$ is also differentiable, it is known as differentiable class κ^1 function. In such an instance, there exists a function $\Theta : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_+$ such that $\xi'(r_0) = \Theta(r_0) > 0$, which is equivalent to $\xi(r_0) = \int_0^{r_0} \Theta(z) dz$. Further, as $\xi(r_0) \rightarrow 1$ as $r_0 \rightarrow \infty$, the function should satisfy $\int_0^\infty \Theta(z) dz = 1$, i.e., functions ξ and Θ can be considered as the probability density function and a cumulative distribution function, respectively, of positive random variables.

Example 2 ([19]). Following is a list of examples of the κ^1 function:

- $\xi(r_0) = 1 - \exp(-r_0)$;
- $\xi(r_0) = \frac{2}{\pi} \arctan(r_0)$;
- $\xi(r_0) = \frac{r_0}{r_0 + \alpha_0}, \alpha_0 > 0$;
- $\xi(r_0) = P(\alpha_0, r_0), \alpha_0 > 0$;
- $\xi(r_0) = I(\alpha_0, \beta_0, r_0)$ with $\alpha_0, \beta_0 > 0$.

2.4. On a lemma

Lemma 1 ([19]). Let $V : \mathbb{R}^p \rightarrow \mathbb{R}_{\geq 0}$ be a continuous, positive definite and radially unbounded function and $\xi \in \kappa^1$ be differentiable in $\mathbb{R} \setminus \{0\}$ such that the trajectories of (1) satisfies:

$$\dot{V}(z) \leq -\frac{1}{(1-\alpha_1)T_c} \frac{(\xi(V(z)))^{\alpha_1}}{\xi'(V(z))} \quad (3)$$

for $z \in \mathbb{R}^p \setminus \{0\}$, $T_c > 0$ and $0 \leq \alpha_1 < 1$; then the origin of (1) is P_DTS in predefined-time T_c . Further, if (3) is an equality then $\sup_{z_0 \in \mathbb{R}^p} T(z_0) = T_c$.

3. Predefined-time stability based on Lyapunov's approach

In this section of the article, a Lyapunov-like theorem for predefined-time stability is presented. The fundamental significance of the theorem is to establish a more general Lyapunov-like sufficient condition for predefined-time stability of the autonomous dynamical system. It will be shown that the Lyapunov-like theorems presented in [19,20,39,40] can be deduced from the given theorem as explained in the Corollaries 1, 2 and 3 and Remark 7.

Theorem 1. Let $V : \mathbb{R}^p \rightarrow \mathbb{R}_{\geq 0}$ be a continuous, positive definite, radially unbounded function and $\xi \in \kappa^1$ be differentiable in $\mathbb{R}_{\geq 0} \setminus \{0\}$. Further, if for any $T_c > 0$ there exist some $\sigma \in \mathbb{R}^q$ such that the time derivative of V along the trajectories of (1) satisfy:

$$\dot{V}(z) \leq -\frac{B(\alpha_0, \beta_0)}{T_c} \frac{(\xi(V(z)))^{1-\alpha_0} (1-\xi(V(z)))^{1-\beta_0}}{\xi'(V(z))} \quad (4)$$

for $z \in \mathbb{R}^p \setminus \{0\}$ and some α_0, β_0 satisfying $0 < \alpha_0 \leq 1, 0 < \beta_0 \leq 1$; then the origin of (1) is $P_D T S$ in predefined-time T_c . Additionally, if (4) is an equality then $\sup_{z_0 \in \mathbb{R}^p} T(z_0) = T_c$.

Proof. Let T_c be any given positive real number. Then there exist some $\sigma \in \mathbb{R}^q$ for which Eq. (4) holds. Also, as $V(z)$ is a continuous, positive definite and radially unbounded function from $\mathbb{R}^p \rightarrow \mathbb{R}_{\geq 0}$ and its time derivative satisfies (4), so, $\dot{V}(z) < 0$ for $z \in \mathbb{R}^p \setminus \{0\}$. Thus, origin of (1) is asymptotically stable [49].

Now, consider $\varphi(t, z_0)$ as a solution of (1) and x be a function from $\mathbb{R}_{\geq 0}$ to $\mathbb{R}_{\geq 0}$ which satisfies:

$$\dot{x} = -\frac{B(\alpha_0, \beta_0)}{T_c} \frac{(\xi(x))^{1-\alpha_0} (1-\xi(x))^{1-\beta_0}}{\xi'(x)} \quad (5)$$

with $V(z_0) \leq x(0)$. Certainly, Eq. (5) can be transferred into the form,

$$\dot{\xi} = -\frac{B(\alpha_0, \beta_0)}{T_c} (\xi(x))^{1-\alpha_0} (1-\xi(x))^{1-\beta_0} \quad (6)$$

$$= U(\xi), \text{ say.} \quad (7)$$

It can be observed from above equation that

(1) $U(0) = 0$;

(2) $\xi U(\xi) \leq 0$ and equals 0 only for $\xi = 0$.

Further, for all $0 \leq v \leq 1$,

$$\begin{aligned} & \int_v^0 \left(\frac{d\xi}{U(\xi)} \right) \\ &= \frac{T_c}{B(\alpha_0, \beta_0)} \int_0^v \xi^{\alpha_0-1} (1-\xi)^{\beta_0-1} d\xi \\ &\leq \frac{T_c}{B(\alpha_0, \beta_0)} \int_0^1 \xi^{\alpha_0-1} (1-\xi)^{\beta_0-1} d\xi \\ &= T_c < \infty. \end{aligned}$$

So, according to the article [2] of Haimo, Eq. (6) is finite-time stable. Let $T(\xi_0) = T(\xi(x(0)))$ be the settling-time. Then, from Eq. (6), it can be obtained that

$$-\frac{B(\alpha_0, \beta_0)}{T_c} \int_0^{T(\xi_0)} dt = \int_{\xi(x(0))}^0 \xi^{\alpha_0-1} (1-\xi)^{\beta_0-1} d\xi,$$

which can be generalized to give

$$T(\xi_0) = \frac{T_c}{B(\alpha_0, \beta_0)} \int_0^{\xi(x(0))} \xi^{\alpha_0-1} (1-\xi)^{\beta_0-1} d\xi.$$

Therefore, it ultimately yields

$$\sup_{\xi_0 \in [0,1)} T(\xi_0) = \sup_{x(0) \in \mathbb{R}_{\geq 0}} T(\xi(x(0))) = T_c.$$

Thus, $\xi(x) = x(t) = 0$ for $t \geq T_c$. Again, by comparison lemma [49] $V(\varphi(t, z_0)) \leq x(t)$. Hence, it is obtained that $V(\varphi(t, z_0)) = 0 = \varphi(t, z_0)$ for $t \geq T_c$, which implies that the origin of the system (1) is finite-time stable and the settling-time function satisfies $\sup_{z_0 \in \mathbb{R}^p} T(z_0) \leq T_c$. Thus, according to Definition 2, the origin of system (4) is $P_D T S$. Further, if Eq. (4) is an equality, then clearly $\sup_{z_0 \in \mathbb{R}^p} T(z_0) = T_c$. $\square$

Remark 4. Consider the function $W : \mathbb{R}^p \rightarrow \mathbb{R}_{\geq 0}$, which is described by [17] as

(i) $W(z) = 0 \Leftrightarrow z = 0$;

(ii) $0 \leq W(z) < 1$;

(iii) $\lim_{\|z\| \rightarrow \infty} W(z) = 1$.

Then, we can formulate Theorem 1 in terms of $W(z)$ as follows:

$$\dot{W}(z) \leq -\frac{B(\alpha_0, \beta_0)}{T_c} W(z)^{1-\alpha_0} (1-W(z))^{1-\beta_0}$$

for $z \in \mathbb{R}^p \setminus \{0\}$ and $W(z) = \xi(V(z))$.

Remark 5. If one has to consider the case of Lyapunov function $V(z)$ in [Theorem 1](#) when, $0 \leq V(z) \leq 1$ then, κ^{-1} function can be considered as a scalar continuous strictly increasing function $\xi : [0, 1] \rightarrow [0, 1]$ satisfying $\xi(0) = 0$ and $\xi(1) = 1$.

Corollary 1. [Lemma 1](#) can be obtained from [Theorem 1](#).

Proof. Considering $1 - \alpha_0 = \alpha_1$ and $\beta_0 = 1$, [Theorem 1](#) reduces to

$$\dot{V}(z) \leq -\frac{B(1 - \alpha_1, 1)}{T_c} \frac{(\xi(V(z)))^{\alpha_1}}{\xi'(V(z))} \quad (8)$$

for $z \in \mathbb{R}^p \setminus \{0\}$, $T_c > 0$. Evidently, $0 \leq \alpha_1 < 1$ as $0 < \alpha_0 \leq 1$. Now,

$$B(1 - \alpha_1, 1) = \frac{\Gamma(1 - \alpha_1) \Gamma(1)}{\Gamma(1 - \alpha_1 + 1)} = \frac{\Gamma(1 - \alpha_1)}{(1 - \alpha_1) \Gamma(1 - \alpha_1)} = \frac{1}{(1 - \alpha_1)}. \quad (9)$$

Combining (8) and (9) the proposed statement is verified. $\square$

Corollary 2. If the Lyapunov function satisfies

$$\dot{V} \leq -\frac{C_{pv}}{T_{pc}} (\alpha_p V^q + b_p)^{K_p}, \quad (10)$$

where $V(z) > 0$; α_p, b_p, K_p, q are positive parameters satisfying $q \in (0, 1]$, $qK_p \in (1, \infty)$ and C_{pv} is given by, $C_{pv} = \frac{1}{(\alpha_p)^q} (b_p)^{\frac{1-qK_p}{q}} \frac{1}{qK_p-1}$, then the system (1) is $P_D T S$ within predefined-time T_{pc} . Now, if one considers $C_{pv} = B(a_1, b_1) (b_p)^{\frac{1-qK_p}{q}} \frac{1}{q(\alpha_p)^q}$ in place of provided C_{pv} , then the result (10), as defined in [39], can be obtained from [Theorem 1](#) by choosing $\xi(r) = I\left(a_1, b_1, \frac{\alpha_p r^q}{\alpha_p r^q + b_p}\right)$, with $a_1 = \frac{1}{q} > 0$, $b_1 = \frac{qK_p-1}{q} > 0$ and $\alpha_0 = \beta_0 = 1$.

Remark 6. Different value of C_{pv} is obtained in [Corollary 2](#) due to the fact that different strategies have been used to bound the settling-time function. Further, the condition $q \in (0, 1]$ used in (10) can be relaxed to $q > 0$ in the above proof.

Now, question arises, can a condition like (10) be obtained for $0 < qK_p \leq 1$ from [Theorem 1](#)? The answer to this query is discussed in the next Corollary.

Corollary 3. If one considers $V(z) \leq 1$ and $K_p = 1$, then a condition like (10) can be obtained from [Theorem 1](#) for $0 < q \leq 1$ as given below:

Case 1: $q = 1$.

In accordance to [Remark 5](#), let us consider $\xi(r) = I\left(a_1, b_1, \frac{\ln(1+c_1 r)}{\ln(1+c_1)}\right)$, where $c_1 = \frac{\alpha_p}{b_p} > 0$, $a_1 = b_1 = 1$. Further, by considering $\alpha_0 = \beta_0 = 1$, one can obtain from [Theorem 1](#) that

$$\dot{V} \leq -\frac{C_{pv}}{T_{pc}} (\alpha_p V + b_p),$$

where, $C_{pv} = \frac{1}{\alpha_p} \ln\left(\frac{\alpha_p + b_p}{b_p}\right)$.

Case 2: $0 < q < 1$.

Consider $\xi(r) = \frac{\int_0^{r^q} \left(\frac{z^{l-1}}{1+c_1 z}\right) dz}{\left(\frac{1}{l}\right) \{ {}_2F_1(1, l; l+1; -c_1) \}}$, where, $l = \frac{1}{q} > 1$. Then, considering $\alpha_0 = \beta_0 = 1$ in [Theorem 1](#) one gets

$$\dot{V} \leq -\frac{C_{pv}}{T_{pc}} (\alpha_p V^q + b_p),$$

where, $C_{pv} = \frac{\left(\frac{1}{l}\right) \{ {}_2F_1(1, l; l+1; -c_1) \}}{qb_p}$.

Remark 7. From [Corollaries 2](#) and [3](#), it is evident that the prior Lyapunov-type theorems on predefined-time stability for autonomous systems defined in [39,40] can be derived from [Theorem 1](#) by appropriately choosing the parameter values and the κ^{-1} functions. Similarly, if we choose $\alpha_0 = \beta_0 = 1$ and $\xi(V(z)) = \frac{2}{\pi} \arctan\left(V^{\frac{\eta}{2}}(z)\right)$, where $\eta \in (0, 1)$, then the Lyapunov form defined in [Theorem 1](#) of [20] can be obtained from [Theorem 1](#).

Remark 8. Several new Lyapunov sufficient conditions for predefined-time stability can be obtained by choosing different values of parameters and κ^{-1} functions. For example, if we fix $\alpha_0 = 1$, $\xi(V(z)) = 1 - \exp(-V(z))$, then [Eq. \(4\)](#) yields

$$\dot{V}(z) \leq -\frac{B(1, \beta_0)}{T_c} \exp(\beta_0 V(z)).$$

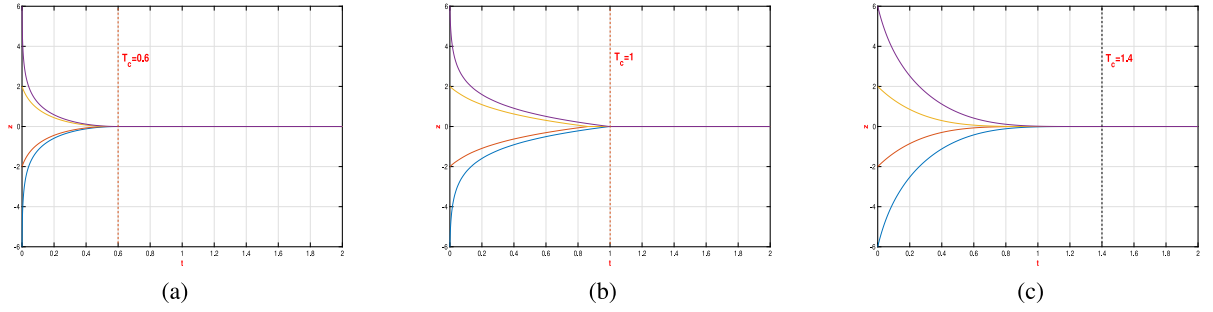


Fig. 1. Trajectories of (11) with different set of parameters, initial conditions ($z_0 = -6, -2, 2, 6$) and predefined-time T_c : (a) $\alpha = 1/2, \beta = 1$ and $T_c = 0.6$ s. (b) $\alpha = 1, \beta = 1$ and $T_c = 1$ s. (c) $\alpha = 1/4, \beta = 1/3$ and $T_c = 1.4$ s.

Example 3. Consider the scalar system defined as

$$\dot{z} = -\frac{B(\alpha_0, \beta_0)}{T_c} \{1 - e^{-|z|}\}^{1-\alpha_0} e^{\beta_0|z|} \text{sign}(z), \quad (11)$$

where $T_c > 0$ and $0 < \alpha_0 \leq 1, 0 < \beta_0 \leq 1$.

Considering the Lyapunov function as $V(z) = |z|$, and the κ^1 differentiable function $\xi(V) = 1 - \exp(-V)$, the state trajectories of (11) satisfy

$$\dot{V}(z) = -\frac{B(\alpha_0, \beta_0)}{T_c} \frac{(\xi(V(z)))^{1-\alpha_0} (1 - \xi(V(z)))^{1-\beta_0}}{\xi'(V(z))}$$

for $z \in \mathbb{R}^p \setminus \{0\}$.

Hence, according to Theorem 1, the origin of the system (11) is P_DTS within predefined-time T_c . Its numerical justification is shown in Fig. 1.

Remark 9. In general, any scalar system of the form

$$\dot{z} = -\frac{B(\alpha_0, \beta_0)}{T_c} \frac{(\xi(V(z)))^{1-\alpha_0} (1 - \xi(V(z)))^{1-\beta_0}}{\xi'(V(z))} \text{sign}(z),$$

for $T_c > 0$ and $0 < \alpha_0 \leq 1, 0 < \beta_0 \leq 1$, is P_DTS with strong predefined-time T_c .

4. Application: To stabilize perturbed first and second order system in predefined-time

4.1. Predefined-time controllers for first-order system

Consider the perturbed first-order system:

$$\dot{z}(t) = f + u + \Delta(z, t), \quad (12)$$

where $z \in \mathbb{R}$ is the system state, $f \in \mathbb{R}$ is known function, $u \in \mathbb{R}$ is the control input and $\Delta(z, t) \in \mathbb{R}$ is the unknown perturbation term including both the external and internal disturbances. Moreover, $\Delta(z, t) \in \mathbb{R}$ is bounded, i.e., $|\Delta(z, t)| \leq \delta$, where, $0 \leq \delta < \infty$ is a known constant. Now, a controller is designed in the following theorem to stabilize the origin of (12) in predefined-time in spite of the presence of the bounded disturbances $\Delta(z, t)$.

Theorem 2. If the controller be designed as

$$u = -f(z) - \left\{ \frac{B(\alpha_0, \beta_0)}{T_c} \frac{(\xi(V(z)))^{1-\alpha_0} (1 - \xi(V(z)))^{1-\beta_0}}{\xi'(V(z))} + \delta \right\} \text{sign}(z), \quad (13)$$

where $0 < \alpha_0 \leq 1, 0 < \beta_0 \leq 1, T_c > 0$, $\xi \in \kappa^1$ is differentiable in $\mathbb{R}_{\geq 0} \setminus \{0\}$ and V is the Lyapunov function, then, the origin of (12) is P_DTS in predefined-time T_c .

Proof. Time derivative of the Lyapunov function candidate $V(z) = |z|$ along the trajectories of the system (12) and (13) provide

$$\begin{aligned} \dot{V}(z) &= \dot{z} \text{sign}(z) \\ &= \left[-\left\{ \frac{B(\alpha_0, \beta_0)}{T_c} \frac{(\xi(V(z)))^{1-\alpha_0} (1 - \xi(V(z)))^{1-\beta_0}}{\xi'(V(z))} + \delta \right\} \text{sign}(z) + \Delta \right] \text{sign}(z) \end{aligned}$$

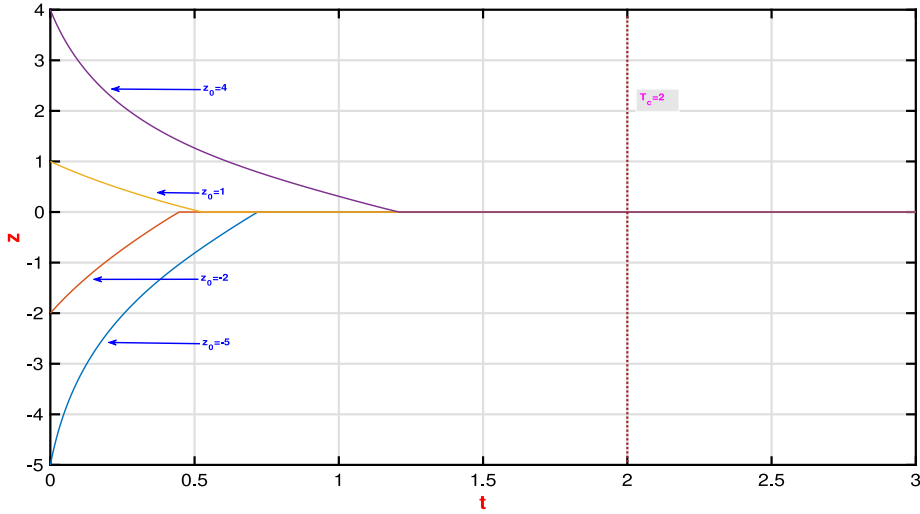


Fig. 2. Trajectories of (12) and (13) with different initial conditions $z_0 = -5, -2, 1, 4$ and predefined-time $T_c = 2$.

$$\begin{aligned}
 &= -\frac{B(\alpha_0, \beta_0)}{T_c} \frac{(\xi(V(z)))^{1-\alpha_0} (1-\xi(V(z)))^{1-\beta_0}}{\xi'(V(z))} - \delta + \Delta \text{sign}(z) \\
 &\leq -\frac{B(\alpha_0, \beta_0)}{T_c} \frac{(\xi(V(z)))^{1-\alpha_0} (1-\xi(V(z)))^{1-\beta_0}}{\xi'(V(z))} - \delta + |\Delta \text{sign}(z)| \\
 &\leq -\frac{B(\alpha_0, \beta_0)}{T_c} \frac{(\xi(V(z)))^{1-\alpha_0} (1-\xi(V(z)))^{1-\beta_0}}{\xi'(V(z))}
 \end{aligned}$$

Hence, according to Theorem 1, the origin of system (12) is $P_D T S$ in predefined-time T_c . $\square$

Remark 10. If we consider $f(z) \equiv 0$ and $\alpha_0 = 1, \beta_0 = 1$, then the controller designed in Theorem 2 is the same as that mentioned in [15]. Further, if we choose $\xi(V(z)) = \xi(|z|) = 1 - \exp(-|z|^m)$ along with $f(z) \equiv 0$ and $\alpha_0 = 1, \beta_0 = 1$, then the above-designed controller (Eq. (13)) is same as that given in the manuscript [14]. Hence, the controller in Theorem 2 is more general as compared to that of [14,15].

Example 4. An example corresponding to the above mentioned methodology for the perturbed system (12) is shown in Fig. 2, where $V(z) = |z|$, $\xi(V(z)) = 1 - \exp(-V(z))$, $f(z) = z^2 + \sin(z)$, $\alpha_0 = 3/4, \beta_0 = 2/3$, with the disturbance $\Delta(z, t) = 1 + \sin(t) \sin(z)$, $\delta = 2$ and the control input u , defined as:

$$u = - \left\{ \frac{B(\alpha_0, \beta_0)}{T_c} (1 - \exp(-V(z)))^{1-\alpha_0} \exp(\beta_0 V) + \delta \right\} \text{sign}(z).$$

4.2. Predefined-time controllers and sliding surface for second-order system

Consider the perturbed second order system:

$$\begin{aligned}
 \dot{z}_1(t) &= z_2(t) \\
 \dot{z}_2(t) &= f + u + \Delta(z_1, z_2, t),
 \end{aligned} \tag{14}$$

where $(z_1, z_2) \in \mathbb{R}^2$ is the system state, $f \in \mathbb{R}$ is known function, u is the control input and $\Delta(z_1, z_2, t) \in \mathbb{R}$ is unknown but bounded disturbances, i.e., $|\Delta(z_1, z_2, t)| \leq \delta$ for some known constant δ , where $0 \leq \delta < \infty$.

Now, being motivated by the sliding surface [8] designed for fixed-time stability, we establish following two theorems which describe the designing of a new sliding surface and a controller for stabilizing the origin of (14) in predefined-time despite the presence of the bounded disturbance $\Delta(z_1, z_2, t)$.

Theorem 3. Construct a $P_D T S S$ for the system (14) as:

$$S = z_2 + \left[|z_2|^2 + 2 \left(\frac{B(\alpha_0, \beta_0)}{T_c} \right)^2 \left(\frac{(\xi(V(z_1)))^{1-\alpha_0} (1-\xi(V(z_1)))^{1-\beta_0}}{\xi'(V(z_1))} \right)^2 \text{sign}(z_1) \right]^{1/2}, \tag{15}$$

where $0 < \alpha_0 \leq 1, 0 < \beta_0 \leq 1, T_c > 0, V(z_1) = |z_1|$, $\xi \in \kappa^1$ is differentiable in $\mathbb{R}_{\geq 0} \setminus \{0\}$. Then, in the case when $S = 0$, the trajectories of system (14) can be directed to the origin in predefined-time T_c .

Proof. When $S = 0$, one can obtain from (15) that

$$z_2 = - \left[|z_2|^2 + 2 \left(\frac{B(\alpha_0, \beta_0)}{T_c} \right)^2 \left(\frac{(\xi(V(z_1)))^{1-\alpha_0} (1 - \xi(V(z_1)))^{1-\beta_0}}{\xi'(V(z_1))} \right)^2 \text{sign}(z_1) \right]^{1/2},$$

Now using the operator $|\bullet|^2$, one can obtain that

$$|z_2|^2 = -|z_2|^2 - 2 \left(\frac{B(\alpha_0, \beta_0)}{T_c} \right)^2 \left(\frac{(\xi(V(z_1)))^{1-\alpha_0} (1 - \xi(V(z_1)))^{1-\beta_0}}{\xi'(V(z_1))} \right)^2 \text{sign}(z_1),$$

which further generalizes to

$$|z_2|^2 = - \left(\frac{B(\alpha_0, \beta_0)}{T_c} \right)^2 \left(\frac{(\xi(V(z_1)))^{1-\alpha_0} (1 - \xi(V(z_1)))^{1-\beta_0}}{\xi'(V(z_1))} \right)^2 \text{sign}(z_1).$$

Again using the operator $|\bullet|^{1/2}$, one can acquire

$$z_2 = - \left(\frac{B(\alpha_0, \beta_0)}{T_c} \right) \left(\frac{(\xi(V(z_1)))^{1-\alpha_0} (1 - \xi(V(z_1)))^{1-\beta_0}}{\xi'(V(z_1))} \right) \text{sign}(z_1),$$

or equivalently, it is

$$z_1(t) = - \left(\frac{B(\alpha_0, \beta_0)}{T_c} \right) \left(\frac{(\xi(V(z_1)))^{1-\alpha_0} (1 - \xi(V(z_1)))^{1-\beta_0}}{\xi'(V(z_1))} \right) \text{sign}(z_1). \quad (16)$$

Now, since (16) is an one-dimensional equation, $V(z_1) = |z_1|$ can act as a Lyapunov function candidate and thus one obtains

$$\dot{V}(z_1) = \dot{z}_1 \text{sign}(z_1). \quad (17)$$

Then, using (16) in (17), one can obtain

$$\dot{V}(z_1) \leq - \frac{B(\alpha_0, \beta_0)}{T_c} \frac{(\xi(V(z_1)))^{1-\alpha_0} (1 - \xi(V(z_1)))^{1-\beta_0}}{\xi'(V(z_1))}.$$

Therefore, according to Theorem 1, $z_1 = 0$ for all $t \geq T_c$ after the $P_D TSS$ surface $S = 0$ is reached by the trajectories of (14). Further, $S = 0$ and $z_1 = 0$ togetherly conclude that $z_2 = 0$. Hence the origin $(z_1, z_2) = (0, 0)$ of (14) is predefined-time stable in predefined-time T_c after the sliding surface is reached. $\square$

Remark 11. One can note that the sliding surface designed in Theorem 2 of the article [15] can be obtained from Theorem 3 by selecting $\alpha_0 = 1, \beta_0 = 1$. Hence, the class of sliding surfaces of the current manuscript is more general than that as defined in [15]. Also, many more sliding surfaces can be obtained corresponding to different choices of the κ^1 function ξ .

In the next theorem, a controller u , based on the analogy of [15], is sketched in such a way that the trajectories of (14) reach the above mentioned sliding surface $S = 0$ in predefined-time.

Theorem 4. Let the $P_D TSS$ surface S be given by (15), $\xi \in \kappa^1$ be twice differentiable in $\mathbb{R}_{\geq 0} \setminus \{0\}$ such that, $\xi'(0) > 0$ and $\xi'' \leq 0$ on $\mathbb{R}_{\geq 0} \setminus \{0\}$. Further, $\xi_s \in \kappa^1$ be differentiable in $\mathbb{R}_{\geq 0} \setminus \{0\}$. Then, for $S \neq 0$, construct a $NP_D TSM$ u as:

$$u = -f - \delta \text{sign}(S) - 2\gamma \Omega \text{sign}(S) - \frac{B(\alpha_1, \beta_1)}{T_s} \frac{(\xi_s(V_s(S)))^{1-\alpha_2} (1 - \xi_s(V_s(S)))^{1-\beta_2}}{\xi'_s(V_s(S))} \text{sign}(S), \quad (18)$$

with

$$\begin{aligned} \Upsilon &= \xi(V(z_1))^{-\alpha_0} (1 - \xi(V(z_1)))^{-\beta_0} \{ (1 - \alpha_0) (1 - \xi(V(z_1))) - (1 - \beta_0) \xi(V(z_1)) \} \\ &\quad - \frac{\xi''(V(z_1)) (\xi(V(z_1)))^{1-\alpha_0} (1 - \xi(V(z_1)))^{1-\beta_0}}{(\xi'(V(z_1)))^2}, \end{aligned} \quad (19)$$

$$\Omega = \left(\frac{B(\alpha_0, \beta_0)}{T_c} \right)^2 \left(\frac{(\xi(V(z_1)))^{1-\alpha_0} (1 - \xi(V(z_1)))^{1-\beta_0}}{\xi'(V(z_1))} \right), \quad (20)$$

where, $0 < \alpha_0 < 1/2, 0 < \beta_0 < 1/2, T_c > 0, V_s(S) = |S|, V(z_1) = |z_1|$ and $0 < \alpha_2 \leq 1, 0 < \beta_2 \leq 1, T_s > 0$. Then the trajectories of (14) can be directed to the sliding surface $S = 0$ within predefined time T_s .

Proof. According to (15), the sliding surface is given by

$$S = z_2 + \left[|z_2|^2 + 2 \left(\frac{B(\alpha_0, \beta_0)}{T_c} \right)^2 \left(\frac{(\xi(V(z_1)))^{1-\alpha_0} (1 - \xi(V(z_1)))^{1-\beta_0}}{\xi'(V(z_1))} \right)^2 \text{sign}(z_1) \right]^{1/2}.$$

Then, considering $V(z_1) = |z_1|$ and differentiating S with respect to time t , one finds

$$\dot{S} = \dot{z}_2 + \frac{|z_2| \dot{z}_2 + 2Y\Omega z_2}{\left[|z_2|^2 + 2 \left(\frac{B(\alpha_0, \beta_0)}{T_c} \right)^2 \left(\frac{(\xi(V(z_1)))^{1-\alpha_0} (1 - \xi(V(z_1)))^{1-\beta_0}}{\xi'(V(z_1))} \right)^2 \text{sign}(z_1) \right]^{1/2}} \quad (21)$$

$$= (f + u + \Delta) + \frac{|z_2| (f + u + \Delta) + 2Y\Theta z_2}{\left[|z_2|^2 + 2 \left(\frac{B(\alpha_0, \beta_0)}{T_c} \right)^2 \left(\frac{(\xi(V(z_1)))^{1-\alpha_0} (1 - \xi(V(z_1)))^{1-\beta_0}}{\xi'(V(z_1))} \right)^2 \text{sign}(z_1) \right]^{1/2}}. \quad (22)$$

Now, substituting the controller u as defined in (18), one can obtain

$$\begin{aligned} \dot{S} &= (\Delta - \delta \text{sign}(S)) - 2Y\Theta \text{sign}(S) - \frac{B(\alpha_2, \beta_2)}{T_s} \frac{(\xi_s(V_s(S)))^{1-\alpha_2} (1 - \xi_s(V_s(S)))^{1-\beta_2}}{\xi'_s(V_s(S))} \text{sign}(S) \\ &+ \frac{|z_2| (\Delta - \delta \text{sign}(S))}{\left[|z_2|^2 + 2 \left(\frac{B(\alpha_0, \beta_0)}{T_c} \right)^2 \left(\frac{(\xi(V(z_1)))^{1-\alpha_0} (1 - \xi(V(z_1)))^{1-\beta_0}}{\xi'(V(z_1))} \right)^2 \text{sign}(z_1) \right]^{1/2}} \\ &- \frac{2Y\Omega (|z_2| \text{sign}(S) - z_2)}{\left[|z_2|^2 + 2 \left(\frac{B(\alpha_0, \beta_0)}{T_c} \right)^2 \left(\frac{(\xi(V(z_1)))^{1-\alpha_0} (1 - \xi(V(z_1)))^{1-\beta_0}}{\xi'(V(z_1))} \right)^2 \text{sign}(z_1) \right]^{1/2}} \\ &- \frac{|z_2| \frac{B(\alpha_2, \beta_2)}{T_s} \frac{(\xi_s(V_s(S)))^{1-\alpha_2} (1 - \xi_s(V_s(S)))^{1-\beta_2}}{\xi'_s(V_s(S))} \text{sign}(S)}{\left[|z_2|^2 + 2 \left(\frac{B(\alpha_0, \beta_0)}{T_c} \right)^2 \left(\frac{(\xi(V(z_1)))^{1-\alpha_0} (1 - \xi(V(z_1)))^{1-\beta_0}}{\xi'(V(z_1))} \right)^2 \text{sign}(z_1) \right]^{1/2}}. \end{aligned}$$

Then, $V_s = |S|$ is a Lyapunov function candidate and differentiating it with respect to time t , one gets

$$\dot{V}_s \leq - \frac{B(\alpha_2, \beta_2)}{T_s} \frac{(\xi_s(V_s(S)))^{1-\alpha_2} (1 - \xi_s(V_s(S)))^{1-\beta_2}}{\xi'_s(V_s(S))}, \text{ for } S \neq 0. \quad (23)$$

Hence, in accordance to Theorem 1, $S = 0$ for all $t \geq T_s$, i.e., the trajectories of the system (14) get to the $P_D TSS$ surface $S = 0$ in predefined-time T_s . Therefore, from Theorems 3 and 4 it can be concluded that the origin $(z_1, z_2) = (0, 0)$ of the system (14), (15) and (18) is $P_D TSS$ in predefined-time $T_c + T_s$. $\square$

Remark 12. The function $V(z_1) = |z_1|$ defined in Theorem 4 is not a Lyapunov function, it is just a function of z_1 constructed in such a way that $\xi(V(z_1))$ can be defined and is only a function of z_1 . Further, from (15) and (18) it can be observed that, one can construct a huge number of $P_D TSS$ and controllers by choosing different $\xi \in \kappa^1$ functions. Also it can be seen from (18) that smaller the value of T_s , larger the energy required to force the state trajectories to the sliding surface. To save the energy, reduce control costs and to enhance practicality are some worthy problems to look after, which we will be considered shortly. Also, by the construction of the controller (18), it is clear that the predefined-time control law (18) is non-singular in the whole state space.

Remark 13. It should be note that the controller in Theorem 4 is designed based on the formation of sliding surface in Theorem 3. Since the sliding surface of the present manuscript is more general than that of the article [15], as mentioned in Remark 11, some more restrictions on the designed controller are to be imposed than that of the controller present in Theorem 2 of [15]. Hence, some restrictions on α_0, β_0 are provided as $\alpha_0, \beta_0 \in (0, 1/2)$ rather than belonging to $(0, 1]$ to cope with this. Moreover, if we give ourselves the liberty to choose $\alpha_0 = \beta_0 = 1$, which can be done if the controller does not pass through the $z_1 = 0$ axis, then further selecting $\alpha_2 = 1, \beta = 1$ and $f \equiv 0$ one can find that the controller defined in Theorem 2 of [15] is same as that of Theorem 4. So, the controller designed in [15] is a special case of the controller of Theorem 4 in some sense.

Example 5 (Synchronization of Φ^6 Duffing Oscillator). Consider the second-order chaotic master system of Φ^6 Duffing oscillator as:

$$\begin{aligned} \dot{z}_{m1} &= z_{m2}, \\ \dot{z}_{m2} &= -az_{m1} - bz_{m2} - cz_{m1}^3 - dz_{m1}^5 + h\cos(\omega t), \end{aligned} \quad (24)$$

and the corresponding slave system as

$$\dot{z}_{s1} = z_{s2}, \quad (25)$$

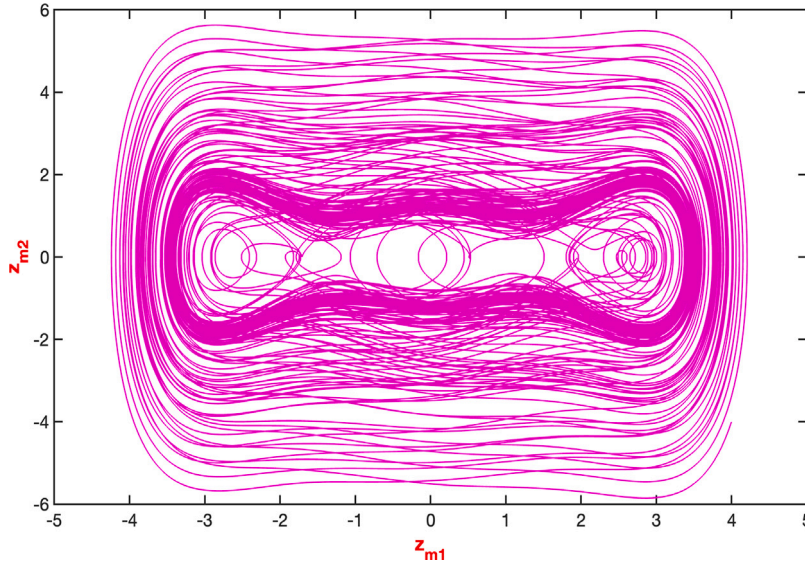


Fig. 3. Chaotic trajectories of (24) with respect to initial condition (4, -4).

$$\dot{z}_{s2} = -az_{s1} - bz_{s2} - cz_{s1}^3 - dz_{s1}^5 + h\cos(\omega t) + u + \Delta(z_{s1}, z_{s2}, t),$$

where $z_{m1}, z_{m2} \in \mathbb{R}$ are the system states of the master system, $z_{s1}, z_{s2} \in \mathbb{R}$ are the system states of the slave system, $u \in \mathbb{R}$ is the control input and $\Delta(z_{s1}, z_{s2}, t)$ is the disturbance in the slave system which is bounded by a non-negative constant δ , i.e., $|\Delta(z_{s1}, z_{s2}, t)| \leq \delta$, where $0 \leq \delta < \infty$. Further, the parameters are given by $a = 1.0, b = 0.01, c = -0.495, d = 0.05, h = 0.78, \omega = 0.55$.

Now, define the tracking error as

$$e_{ms1} = z_{s1} - z_{m1}, e_{ms2} = z_{s2} - z_{m2}.$$

Then, combining (24) and (25), one can have

$$\begin{aligned} \dot{e}_{ms1} &= e_{ms2}, \\ \dot{e}_{ms2} &= -ae_{ms1} - be_{ms2} - c(z_{s1}^3 - z_{m1}^3) - d(z_{s1}^5 - z_{m1}^5) + u + \Delta(z_{s1}, z_{s2}, t). \\ &= f + u + \Delta(z_{s1}, z_{s2}, t), \text{ (say)} \end{aligned} \quad (26)$$

According to Theorems 3 and 4, the origin $(e_{ms1}, e_{ms2}) = (0, 0)$ of system (26) is $P_D T S$ in predefined-time $T_c + T_s$, i.e., $z_{s1} = z_{m1}, z_{s2} = z_{m2}$ for all $t \geq T_c + T_s$.

Now, for numerical verification let us consider $\xi(V(e_{ms1})) = \frac{V(e_{ms1})}{1+V(e_{ms1})}$ and $\xi_s(V_s(S)) = \frac{V_s(S)}{1+V_s(S)}$, where $V(e_{ms1}) = |e_{ms1}|$ and $V_s(S) = |S|$. Without any loss of generality, let us denote $V(e_{ms1})$ by V and $V_s(S)$ by V_s , substituting these values in (15) and (18), one can obtain the $NP_D T S M$ surface $S = 0$ and the $NP_D T S M$ controller u , respectively, as:

$$S = e_{ms2} + \left[|e_{ms2}|^2 + 2 \left(\frac{B(\alpha_0, \beta_0)}{T_c} \right)^2 (V^{1-\alpha_0} (V+1)^{\alpha_0+\beta_0})^2 \text{sign}(e_{ms1}) \right]^{1/2}. \quad (27)$$

$$u = -f - \delta \text{sign}(S) - 2Y\Omega \text{sign}(S) - \frac{B(\alpha_1, \beta_1)}{T_s} (V_s^{1-\alpha_1} (V_s+1)^{\alpha_1+\beta_1}) \text{sign}(S), \quad (28)$$

where,

$$Y\Omega = \left(\frac{B(\alpha_0, \beta_0)}{T_c} \right)^2 V^{1-2\alpha_0} (1+V)^{2\alpha_0+2\beta_0-1} \{1 - \alpha_0 + V + \beta_0 V\}.$$

Fig. 5 validates the suggested strategy (see also Figs. 3, 4, 6 and 7).

5. Conclusion

In the present work, a Lyapunov-like sufficient condition for $P_D T S$ of the dynamical system has been proposed by using a special kind of function known as the κ^1 class function. Moreover, it is shown that the proposed theorem can be considered as a generalization of the previously established many Lyapunov theorems on $P_D T S$ for autonomous systems [19,20,39,40].

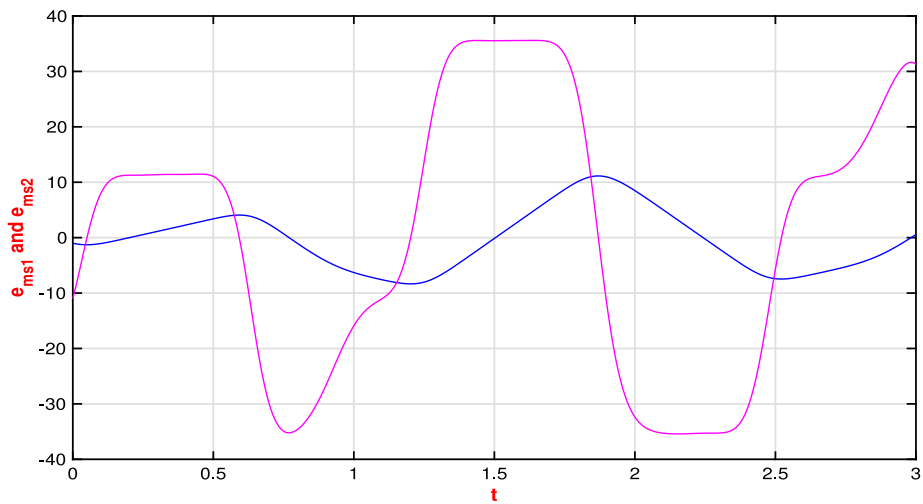


Fig. 4. Error trajectories of (26) with initial condition $(-1, -11)$, without involvement of the controller u and considering $\Delta(z_{s1}, z_{s2}, t) = 1 + \sin(t)\cos(z_{s2})$, $\delta = 2$. Blue trajectory corresponds e_{ms1} and magenta trajectory represents e_{ms2} .

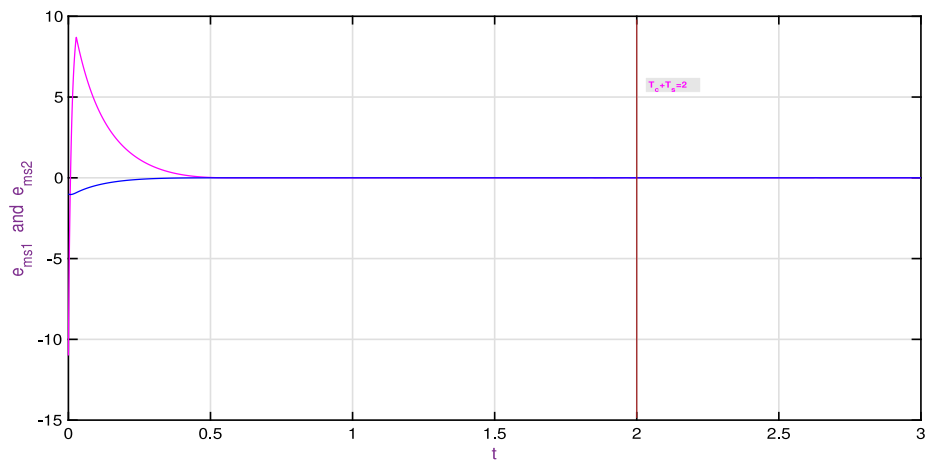


Fig. 5. Error trajectories of (26) with initial condition $(-1, -11)$, with involvement of the controller u and considering $\Delta(z_{s1}, z_{s2}, t) = 1 + \sin(t)\cos(z_{s2})$, $\delta = 2$, $\alpha_0 = 1/4$, $\beta_0 = 1/3$, $\alpha_1 = 1/8$, $\beta_1 = 2/3$, $T_c = 1$, $T_s = 1$. Moreover, blue trajectory represents e_{ms1} and magenta trajectory is for e_{ms2} .

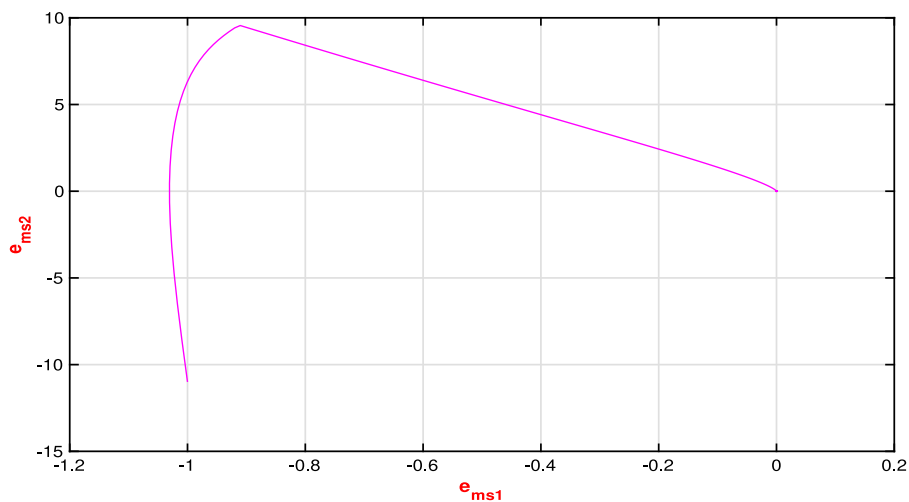


Fig. 6. Phase plot of (26).

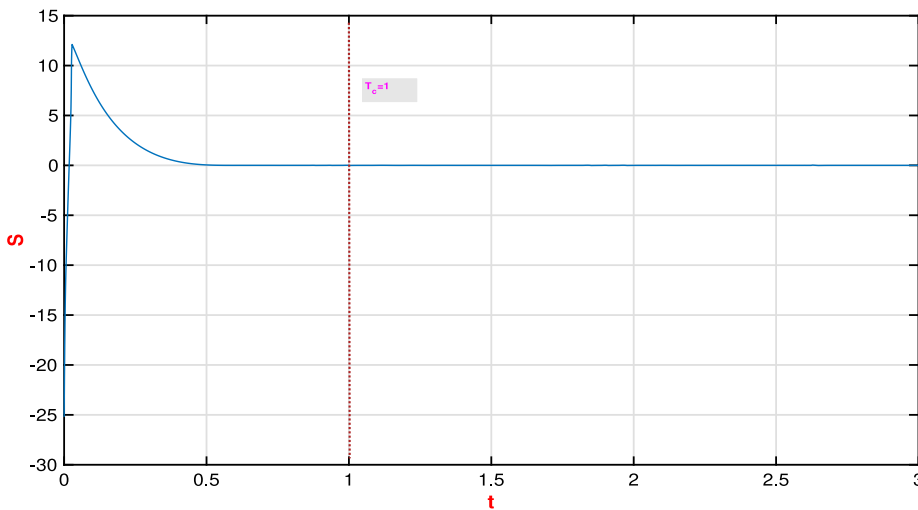


Fig. 7. Evaluation of sliding surface (27) corresponding to the error system (26) with $\alpha_0 = 1/4$, $\beta_0 = 1/3$ and $T_c = 1$.

Subsequently, non-singular predefined-time controllers for the first and second-order systems are developed along with a predefined-time terminal sliding surface for the second-order system. Also, synchronization of Φ^6 Duffing oscillator in predefined-time has been established in presence of external disturbances, along with some more examples to justify the truthfulness of the proposed scheme numerically.

Due to the presence of delay in any practical model, P_DTS of the delayed nonlinear system will be a fruitful topic to investigate in future. Further, the concept of P_DTS along with designing novel predefined-time sliding mode surfaces for the higher-order system will be considered in future. Also, P_DTS of dynamical systems by incorporating time-varying gains [51,52] will be an interesting area to work with, keeping in mind the limitations of it [53]. Moreover, as P_DTS is closely related to tunable parameters; we will be focusing on the synchronization of complex and neural networks in prespecified-time T_0 via a smooth terminal sliding mode control approach by considering bounded disturbances into account, where T_0 is independent of initial conditions and parameters of the system [54,55].

CRediT authorship contribution statement

Arnab Mapui: Conceptualization, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Md Arzoo Jamal:** Investigation, Supervision, Validation, Writing – review & editing. **Santwana Mukhopadhyay:** Investigation, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgements

Authors are thankful to the editor and the anonymous reviewers for their constructive comments and suggestions which have improved the quality of the paper. One of the authors, Arnab Mapui, thankfully acknowledges the full financial assistance of the JRF Fellowship (Roll Number DL0116211074, Reference number 191620152093) by the University Grant Commission (UGC), India, to carry out the present work.

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